

X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-10-01

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:

- CSP Partial cross section data
- CST Temperature dependent cross section
- ETA Average neutron yield per nonelastic event
- NU Fission-neutron yield, nu-bar
- TKE Total kinetic energy
- SIG Cross section data given with limit as DATA and DATA-MIN
- SIG Production cross section, where product coded as SF4 or in DATA (ELEM/MASS)
- PFNS Prompt-fission neutron energy spectra

2. Plotting partial cross sections from local EXFOR and remote ENDF databases

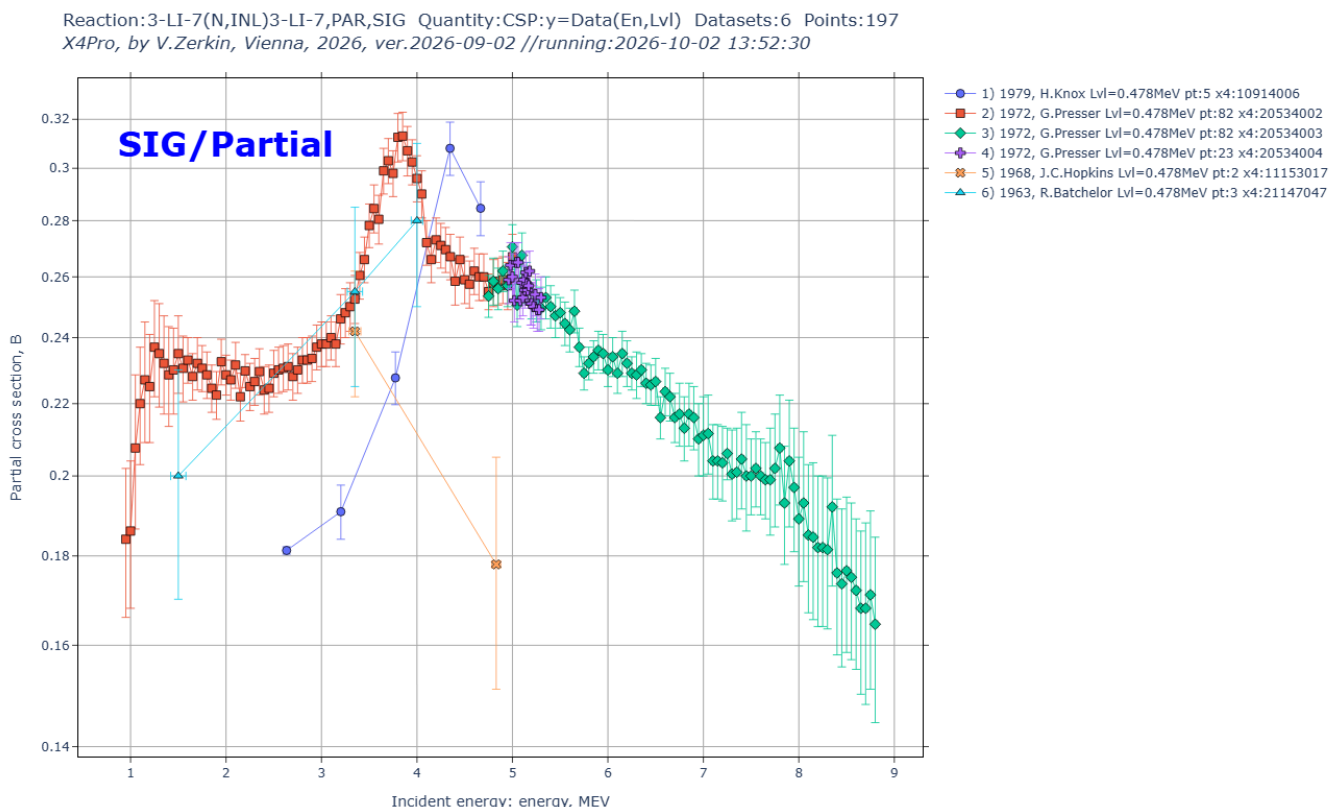
- CSP vs ENDF MT:51/MF:3,33 - Production of a neutron, with residual in the 1st excited state
- NU vs ENDF MT452,455,456/MF:1,31 - Average number of total/delayed/prompt neutrons released per fission event
- ETA vs ENDF MT:452,18,102/MF:1,3,31,33 - Average neutron yield per nonelastic event: $\eta \pm \Delta\eta$
- PFNS vs ENDF - Prompt-fission neutron energy spectra (ratio to Maxwellian: $T=1.32\text{MeV}$ by default)

Code reac1.py help:

```
Program: reac1.py, ver.2026-08-14
Author: V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
* computational data are presented as function y=f(x1,x2,...)
  where f is measured quantity, x1 are independent variables:
    EN - incident energy,          E2 - energy of outgoing particle,
    ANG - angle of outgoing particle,  EVL - energy level, etc.
Algorithm:
* user defines EXFOR Reaction-code for data search, x-variable by number
* selects x-variable by number (default: x1)
* other variables will be used as parameter for grouping of data points
Usage:
$ python [{flag}] x4update.py [{option}|reaction]
* flag: see all Python flags: $ python --help
  -B          don't write .pyc files on import
* option:
  -help       print this help-text and exit (also --help)
  -o:<file>   output file name
  -x:x?       set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
  -x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
  -x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
  -x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
  -nogrp      avoid grouping datasets by reaction-codes on the plot
  -fx:<num>    set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
  -fy:<num>    set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
plotting:
  -xlog       logarithmic scale of X-Axis
  -ylog       logarithmic scale of Y-Axis
  -lines      connect points on the plot by lines
  -sym        draw border of a symbol of a data point on the plot
  -annot:x,y,txt annotation on top of the plot;
              x and y should be given in units of the plot;
              text can include html tags: <b>, <sup>
* reaction:
  <reaction> full EXFOR Reaction-code, e.g. "8-0-16(N,EL)8-0-16,,DA"
```

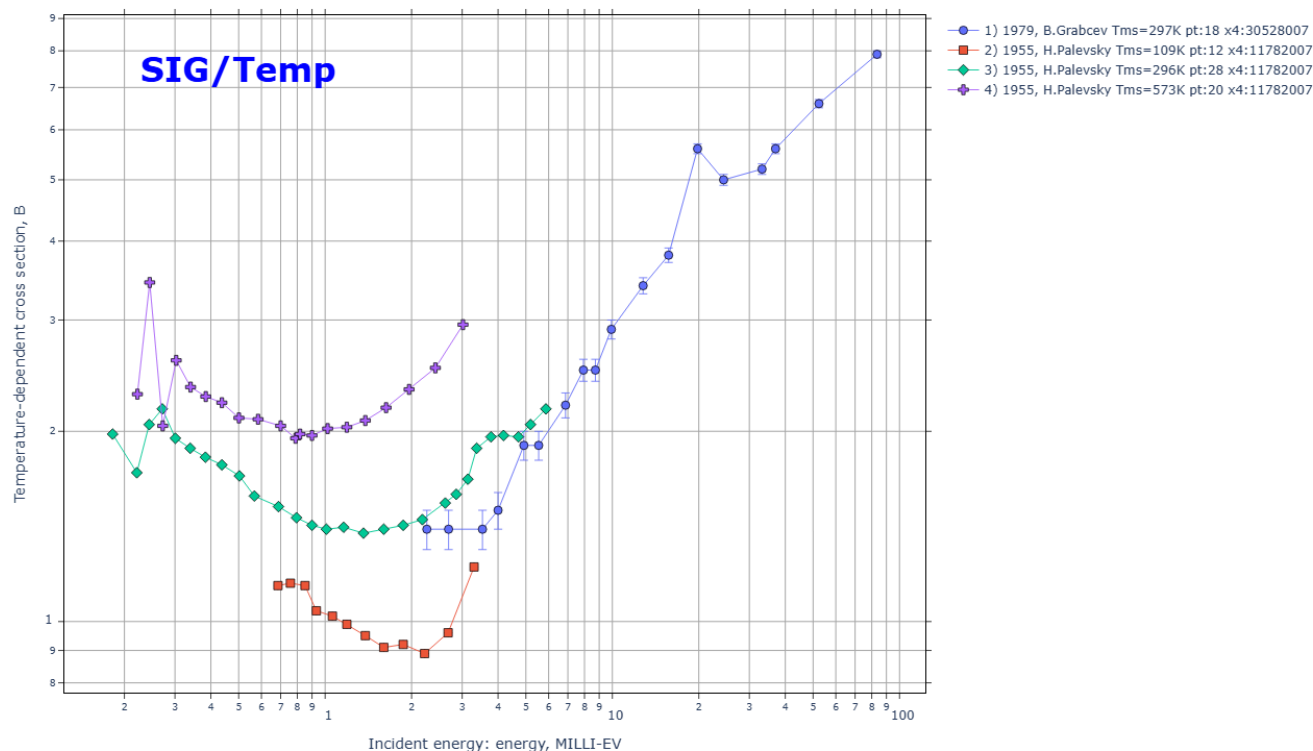
part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data



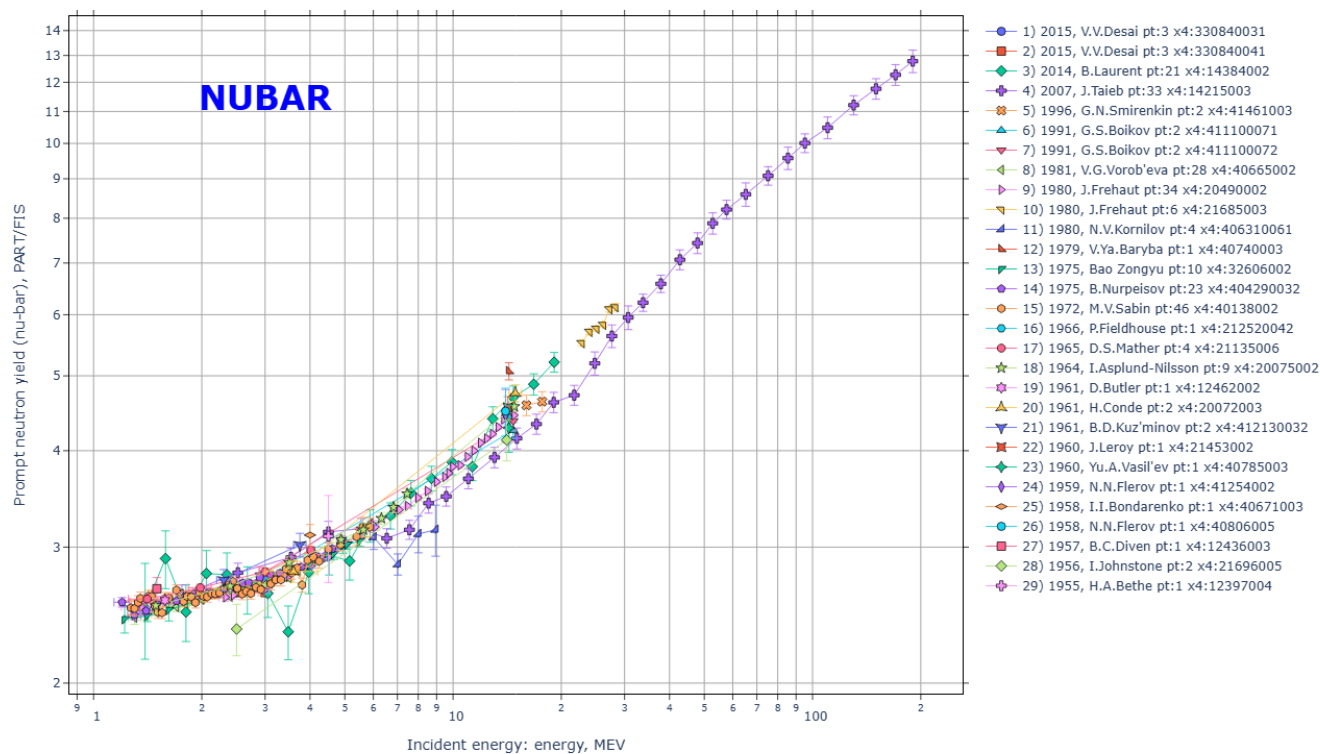
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section

Reaction: 82-PB-0(N,TOT),,SIG/TMP Quantity: CST:y=Data(En,Tms) Datasets: 4 Points: 78
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:19



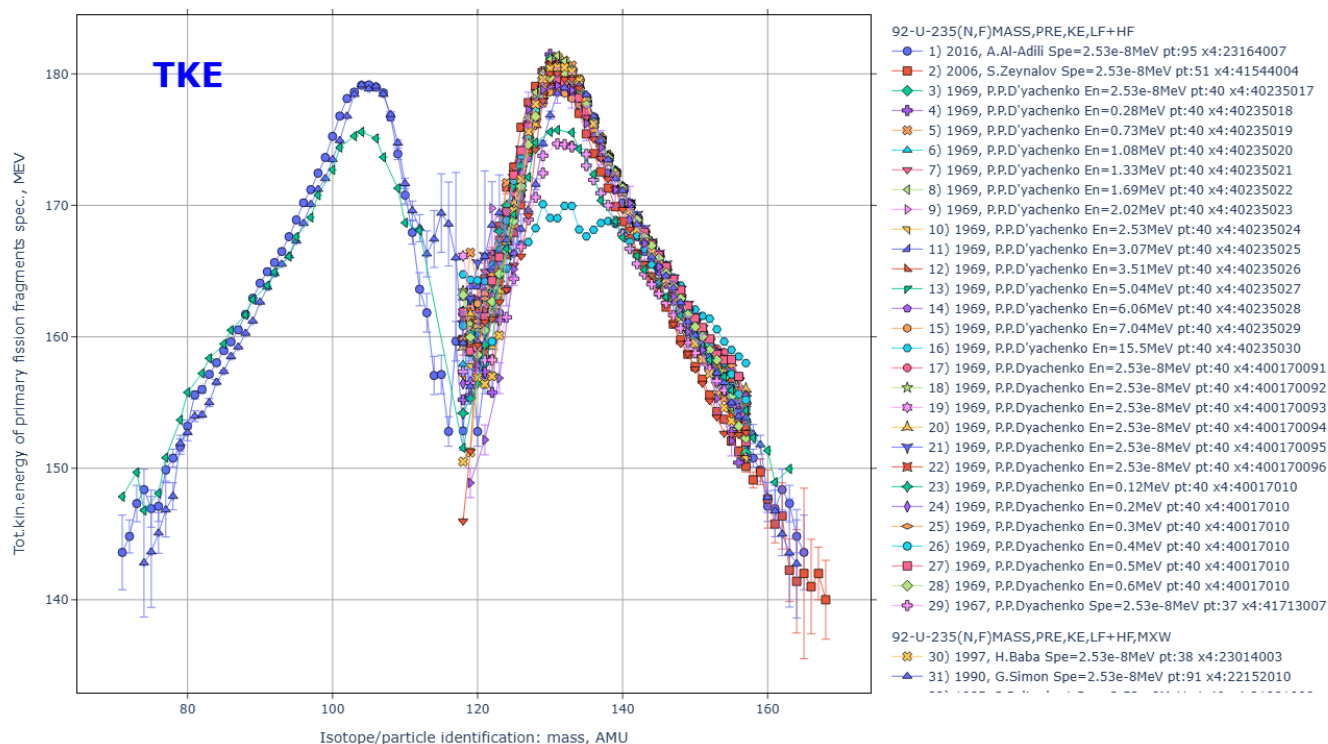
3) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar

Reaction: 92-U-238(N,F),PR,NU Quantity: NU:y=Data(En) Datasets: 29 Points: 246
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:21



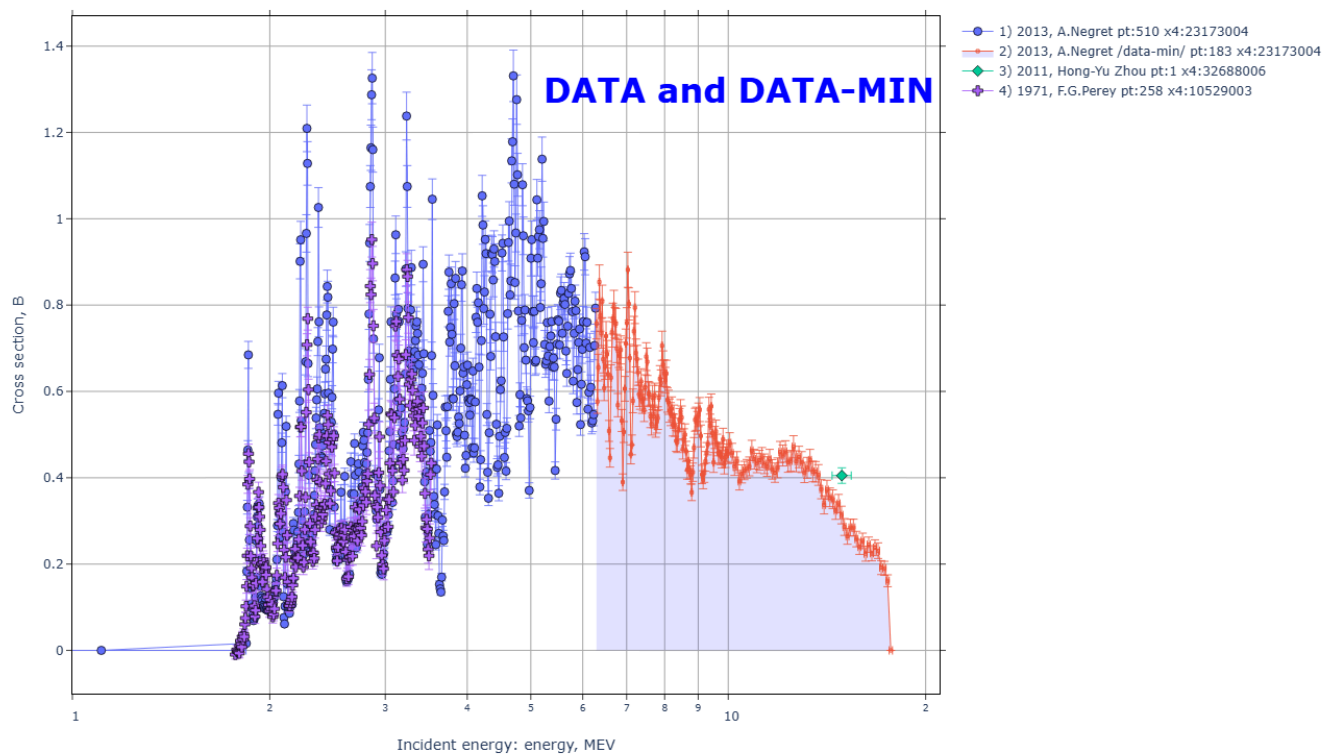
4) [part1-8-reac/out00/tke1.html.png](#) TKE Total kinetic energy

Reaction: 92-U-235(N,F)MASS,PRES,KE,LF+HF Quantity: E: y=Data(Spe,Zam) Datasets: 34 Points: 1467
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:24



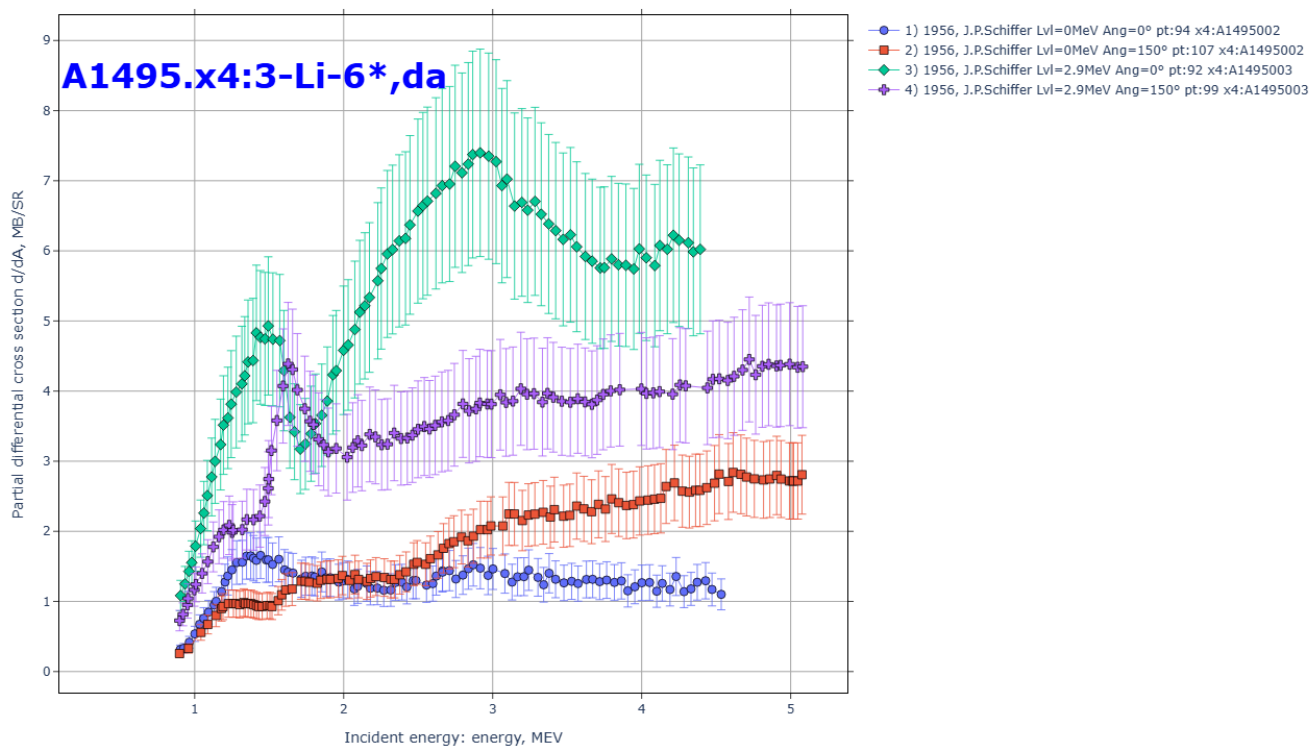
5) [part1-8-reac/out00/si28nn.html.png](#) SIG Cross section data given with limit as DATA and DATA-MIN

Reaction: 14-SI-28(N,INL)14-SI-28,,SIG Quantity: CS: y=Data(En) Datasets: 4 Points: 952
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:33

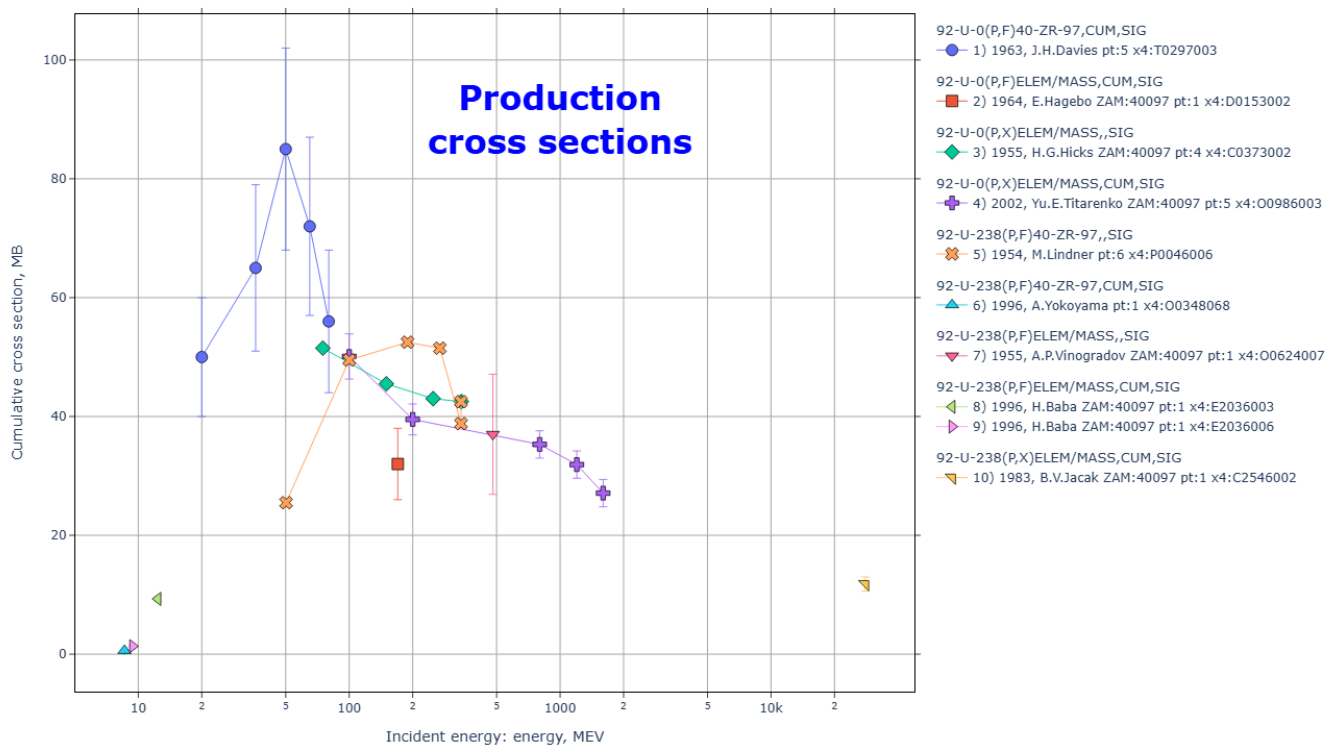


6) part1-8-reac/out00/a1495-li6.html.png Data from EXFOR:"A1495" & Reaction: "3-LI-6*DA"

Reaction:3-LI-6(HE3,P)4-BE-8,PAR,DA Quantity:DAP:y=Data(En,Lvl,Ang) Datasets:4 Points:392
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:35

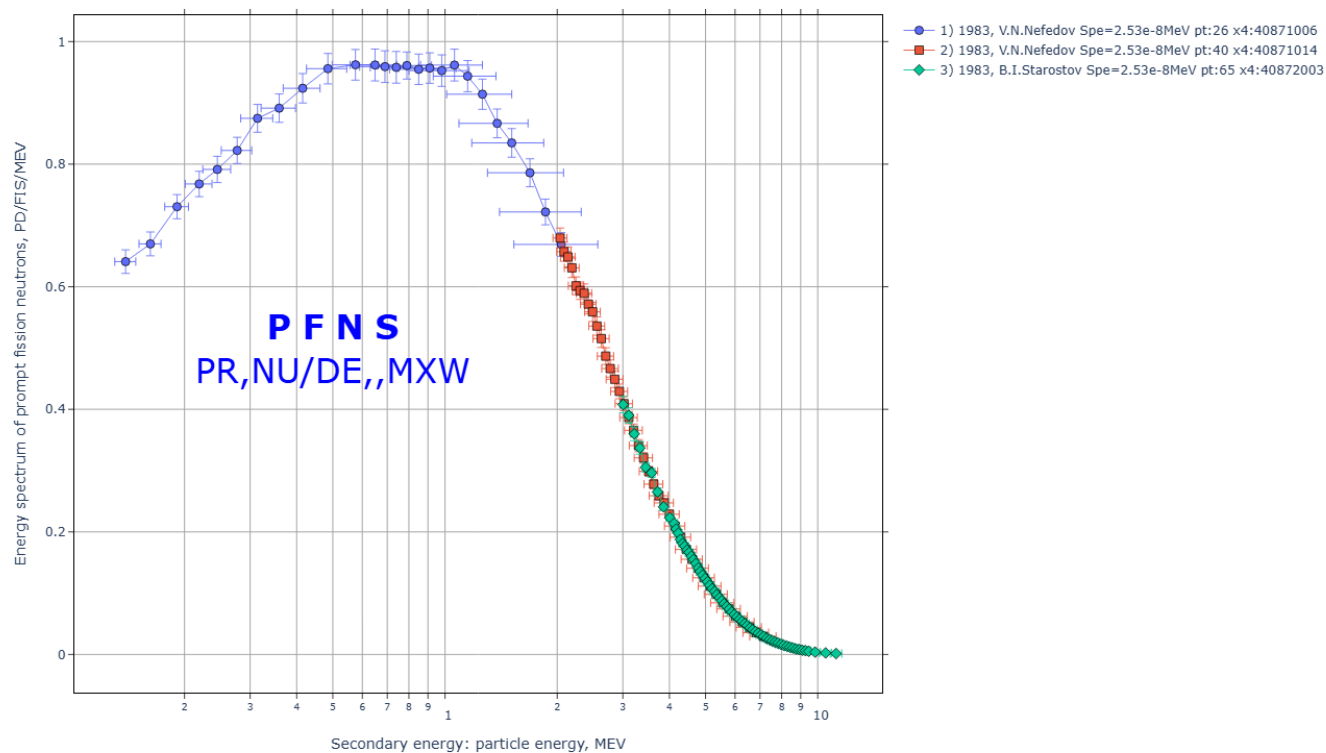
**7) part1-8-reac/out00/sig1prod.html.png** Production cross section coded as SF4 and ELEM/MASS

Reaction:92-U-0(P,F)40-ZR-97,CUM,SIG Quantity:CS:y=Data(En) Datasets:10 Points:26
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:44



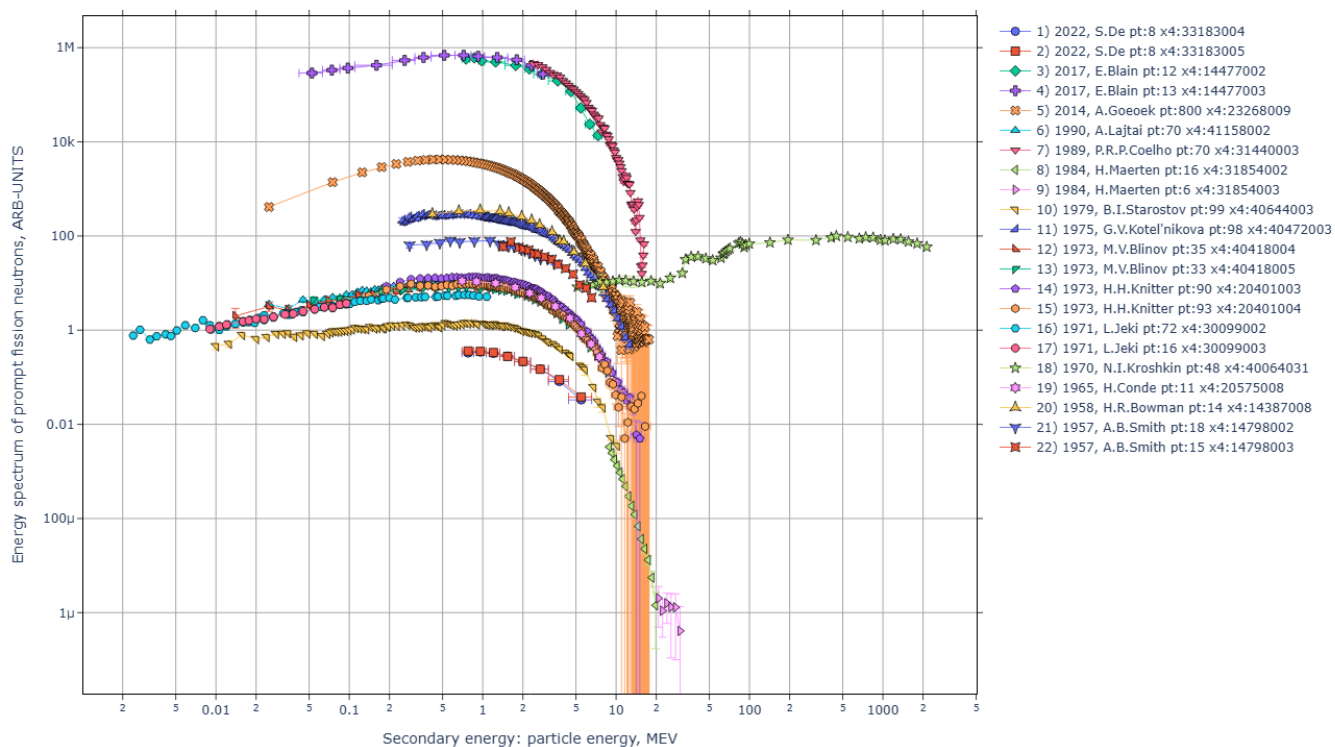
8) [part1-8-reac/out00/pfns1.html.png](#) PFNS Prompt-fission neutron energy spectra

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:46

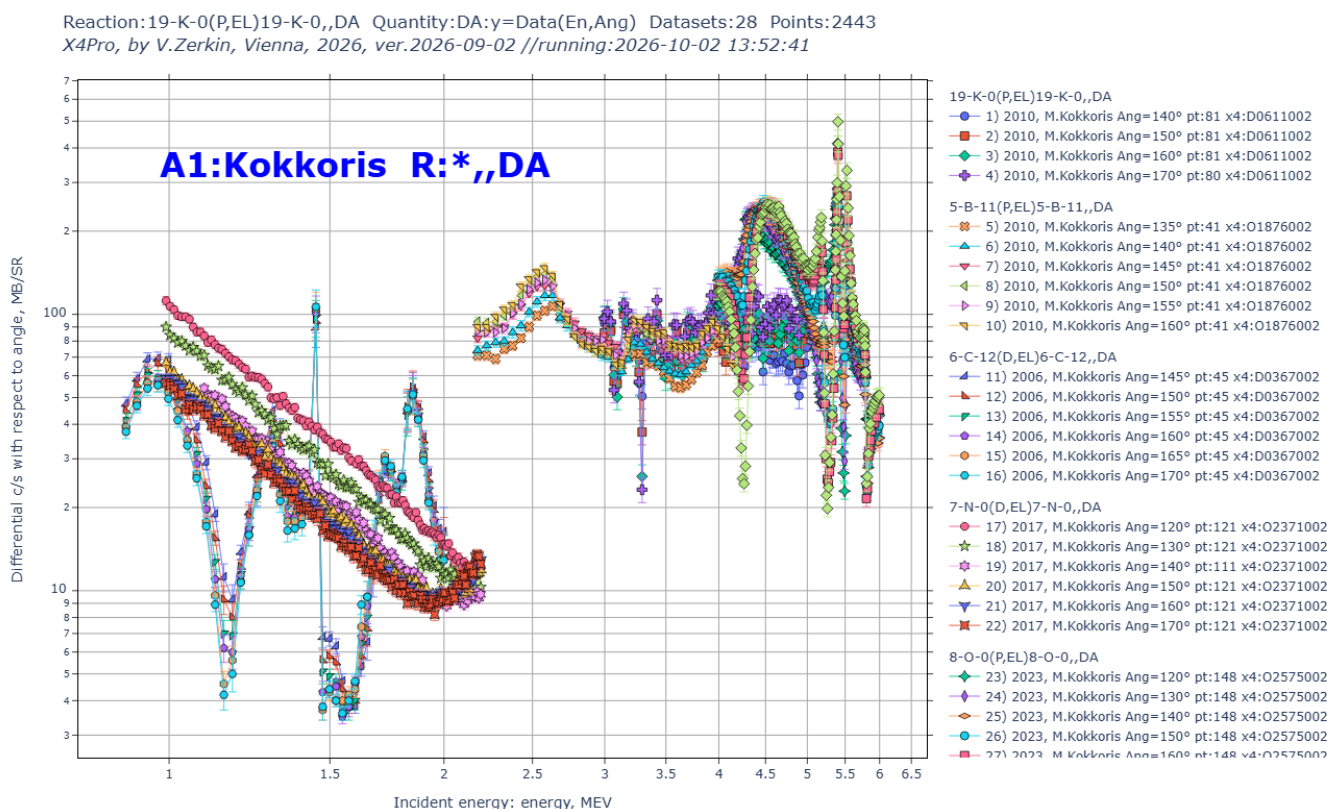


9) [part1-8-reac/out00/pfns4.html.png](#) Energy spectrum of prompt fission neutrons: relative data

Reaction: 98-CF-252(0,F),PR,NU/DE,,REL Quantity: MFQ:y=Data(E2) Datasets: 22 Points: 1645
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:54

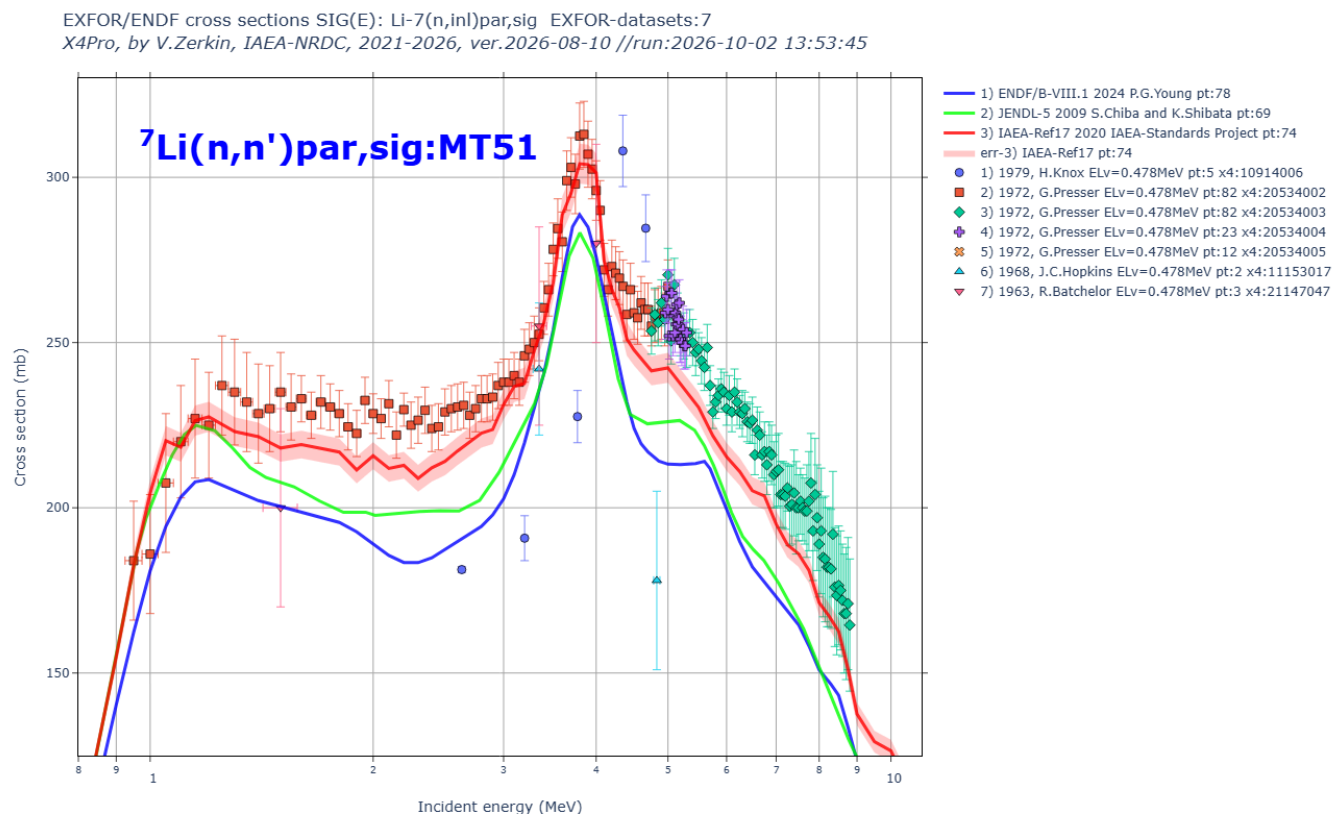


10) part1-8-reac/out00/Kokkoris.html.png Data by "Kokkoris" & Reaction: "*,DA"



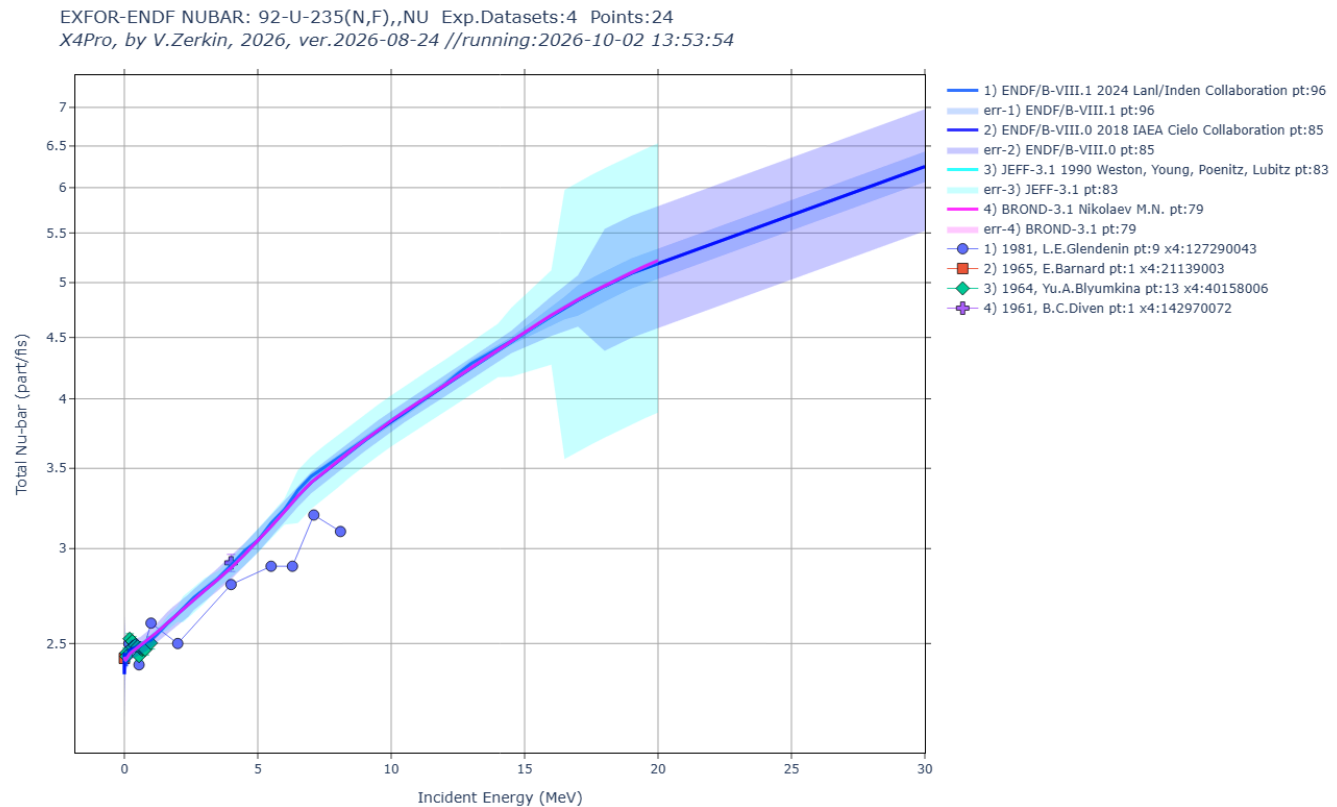
part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

11) part2-8-sig1par/out00/Li7ninl.html.png Production of a neutron, with residual in the 1st excited state

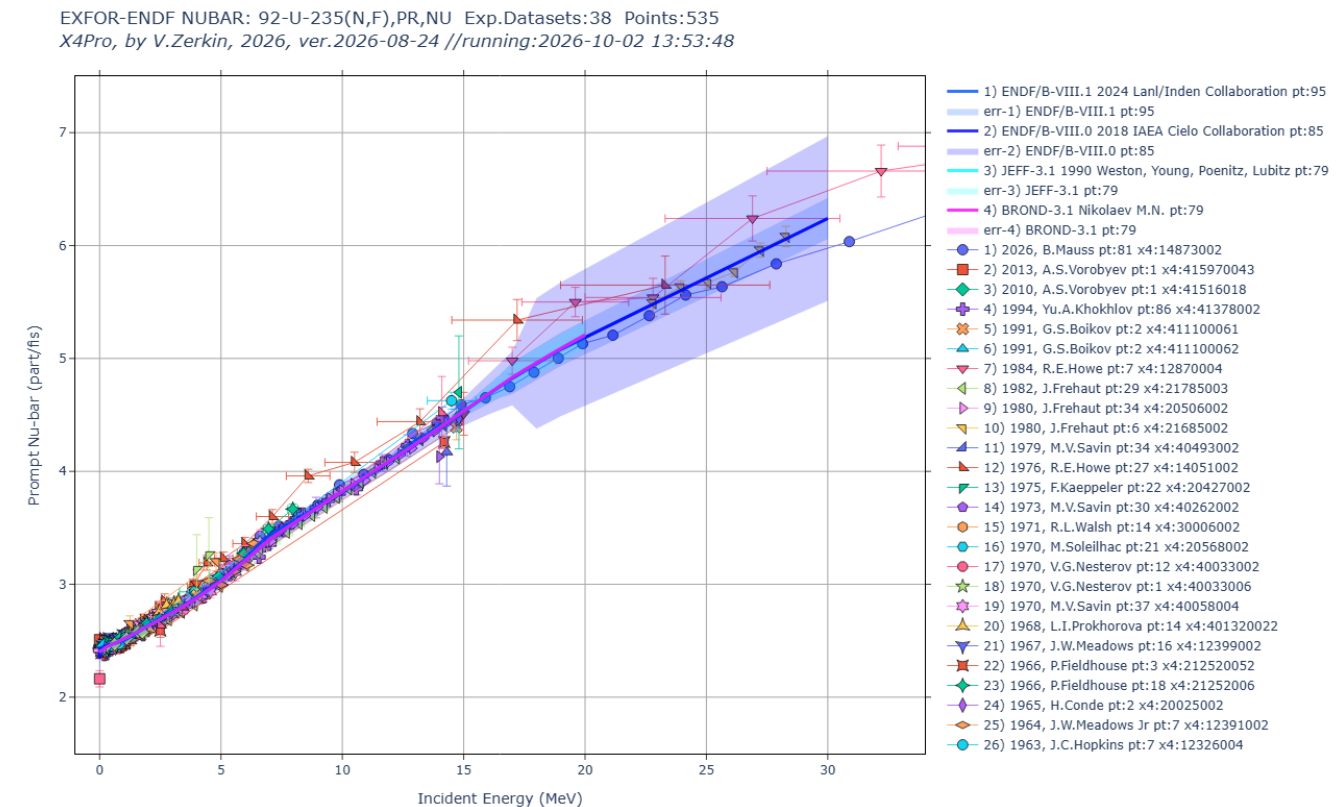


part2-9-nubar/ EXFOR/ENDF NUBAR - average number of neutrons per fission

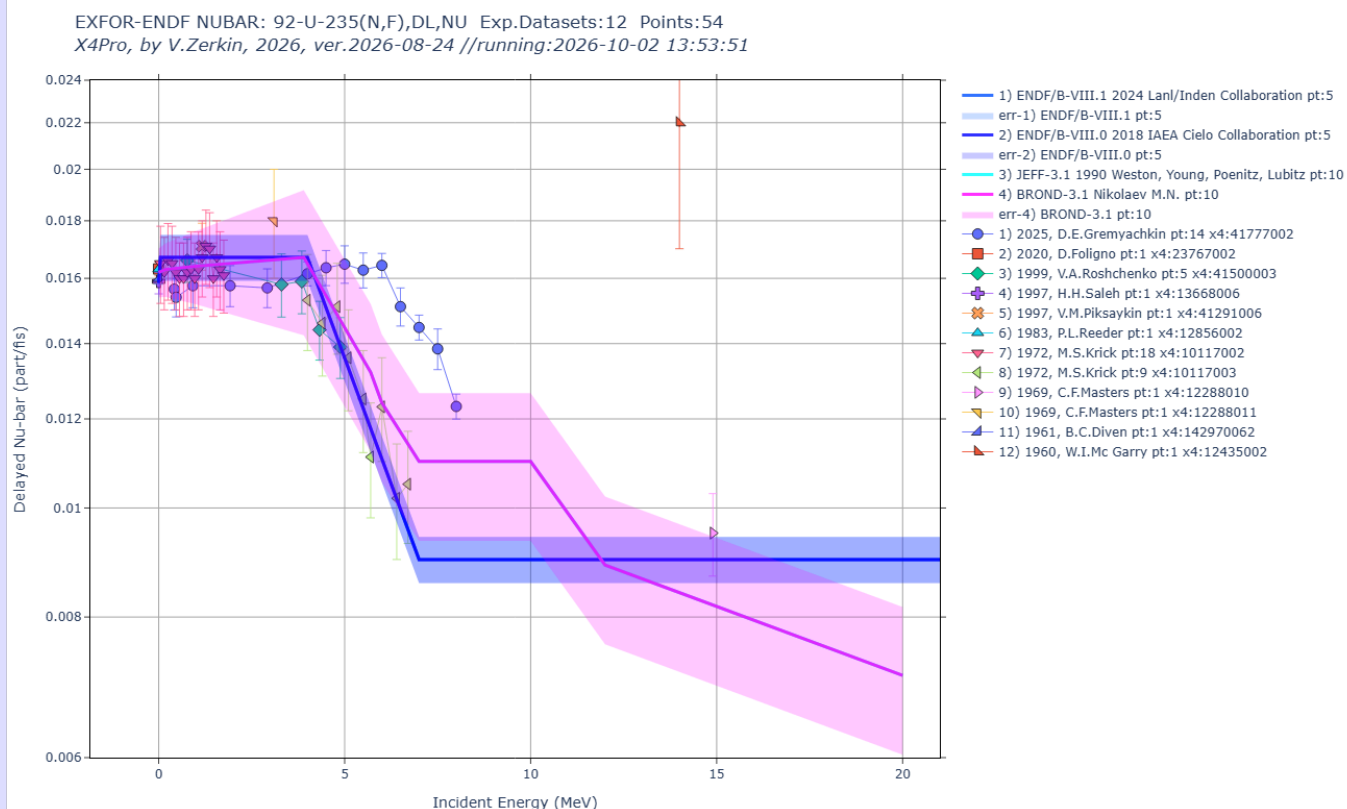
12) part2-9-nubar/out00/U235nf_-nu.html.png Total neutron yield (nu-bar)



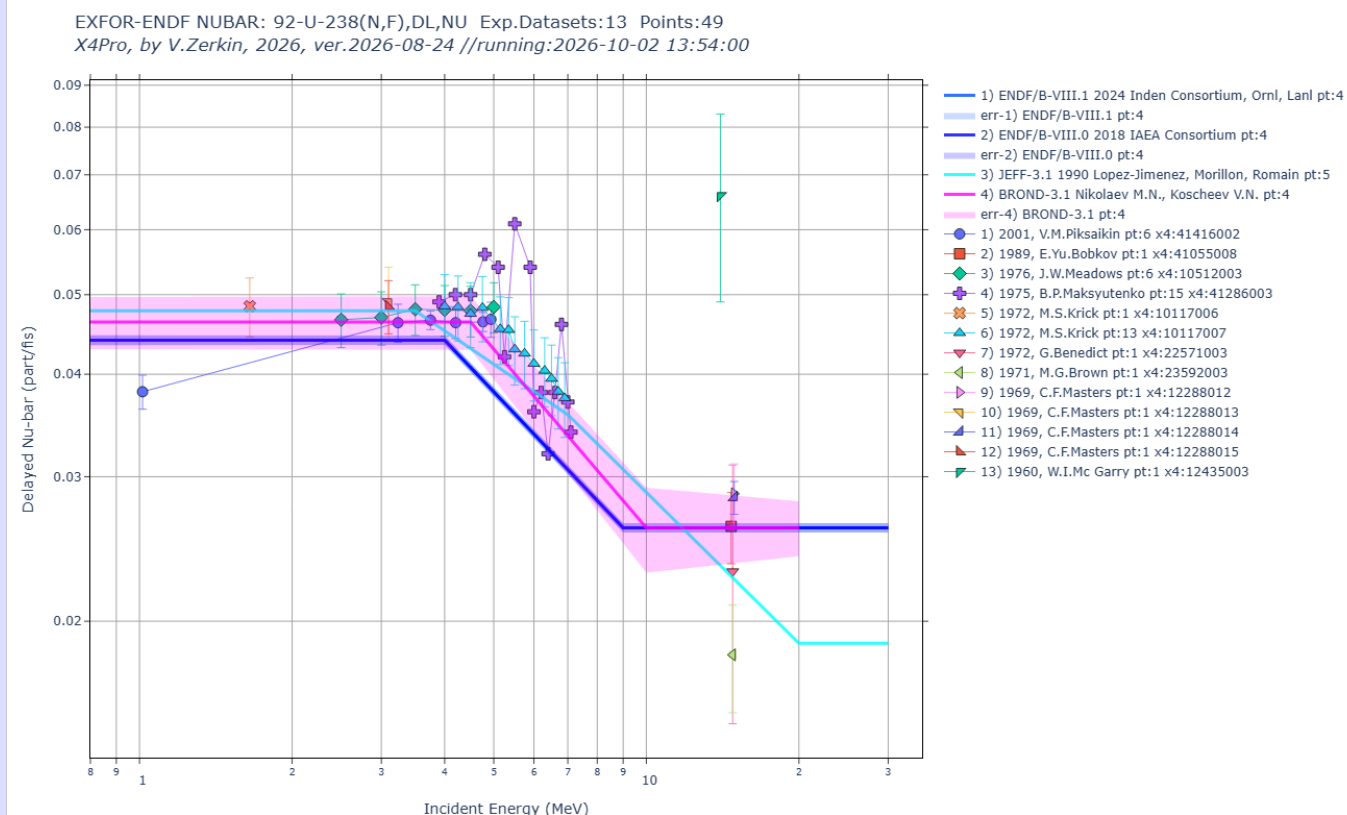
13) part2-9-nubar/out00/U235nf_pr-nu.html.png Prompt neutron yield (nu-bar)



14) [part2-9-nubar/out00/U235nf_dl-nu.html.png](#) Delayed neutron yield



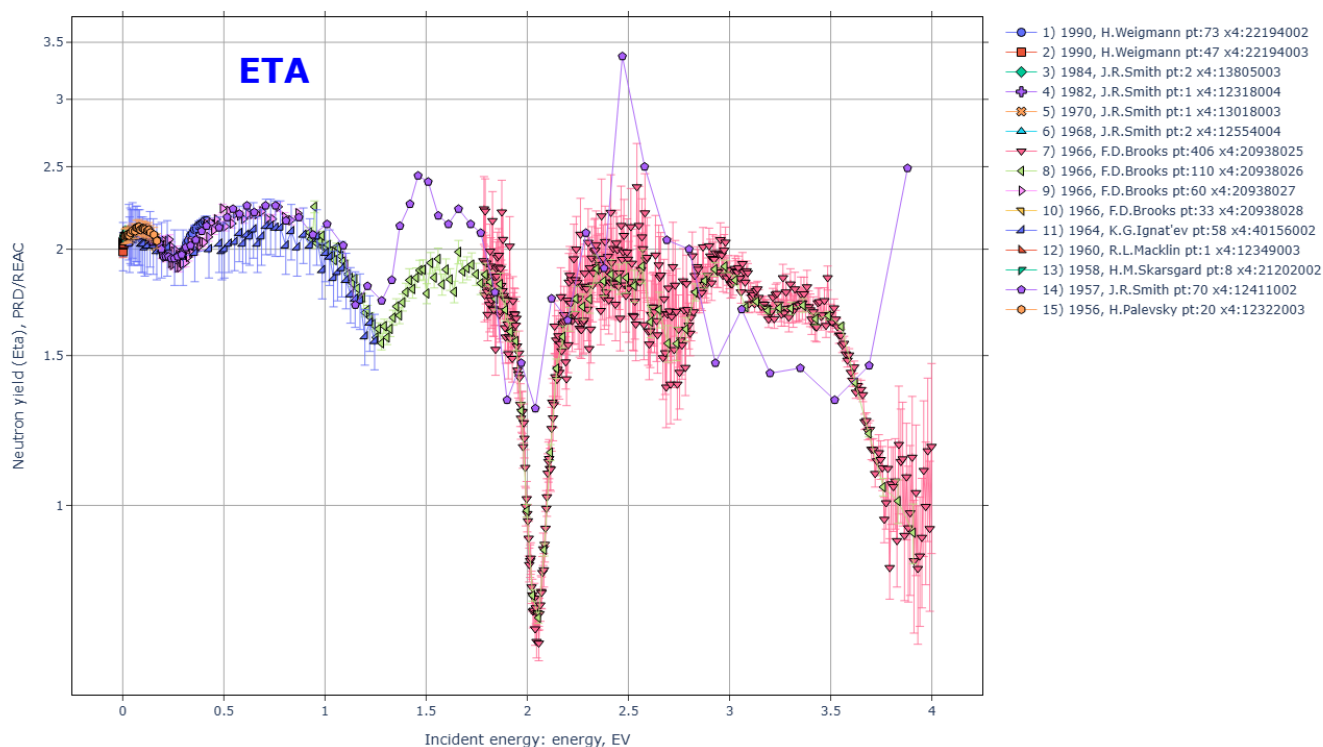
15) [part2-9-nubar/out00/U238nf_dl-nu.html.png](#)



part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

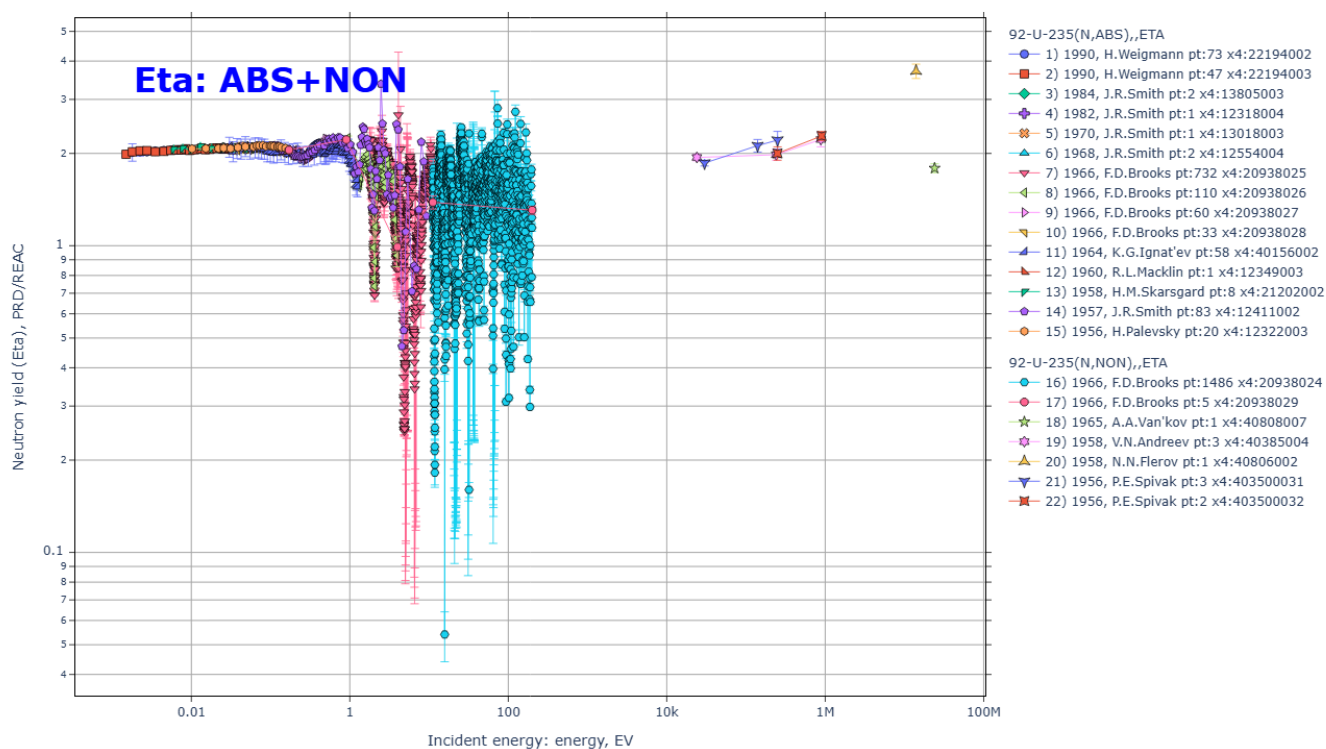
16) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

Reaction:92-U-235(N,ABS),,ETA Quantity:NU:y=Data(En) Datasets:15 Points:892
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-08-28 13:26:55

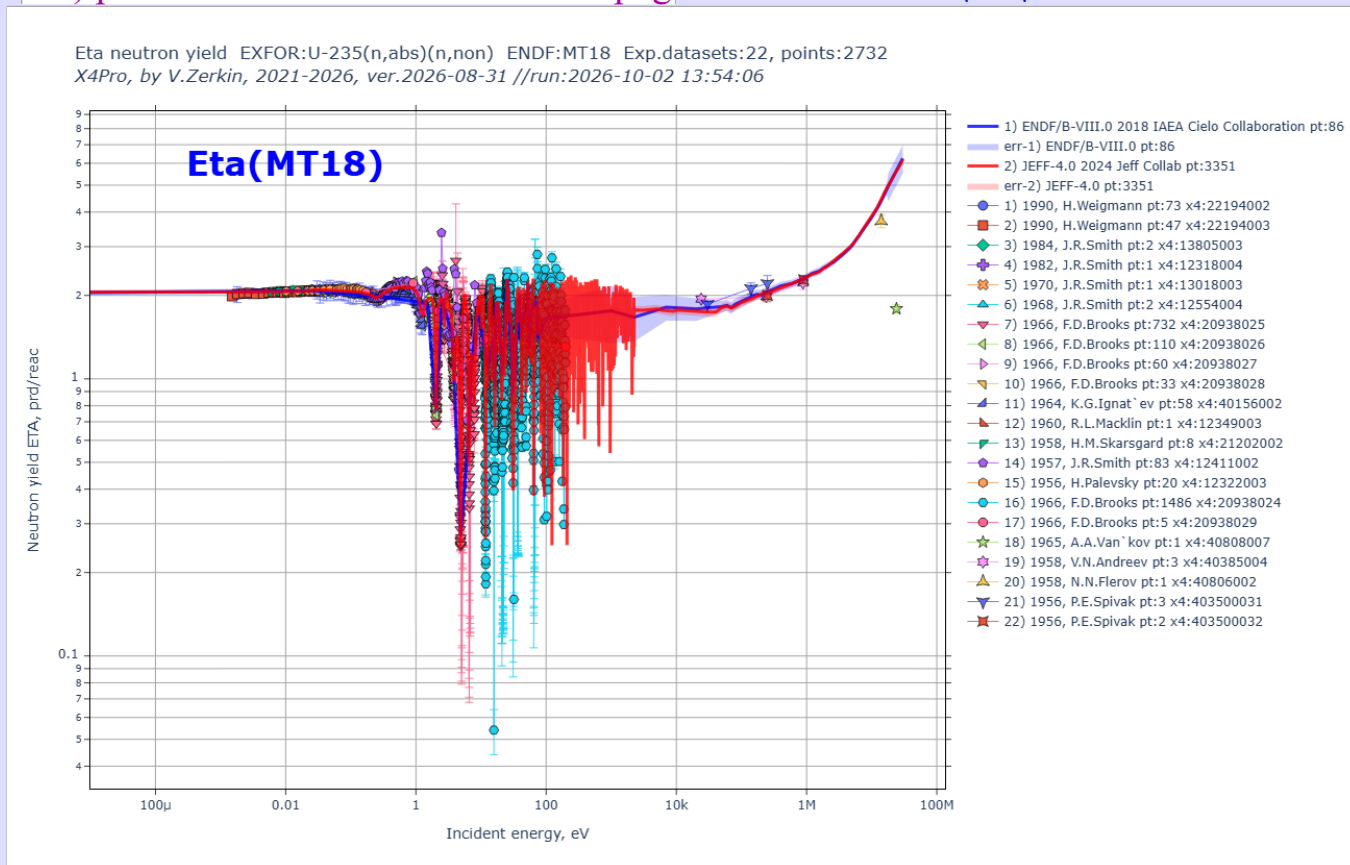


17) part1-8-reac/out00/eta2.html.png ETA: (N,ABS)+(N,NON)

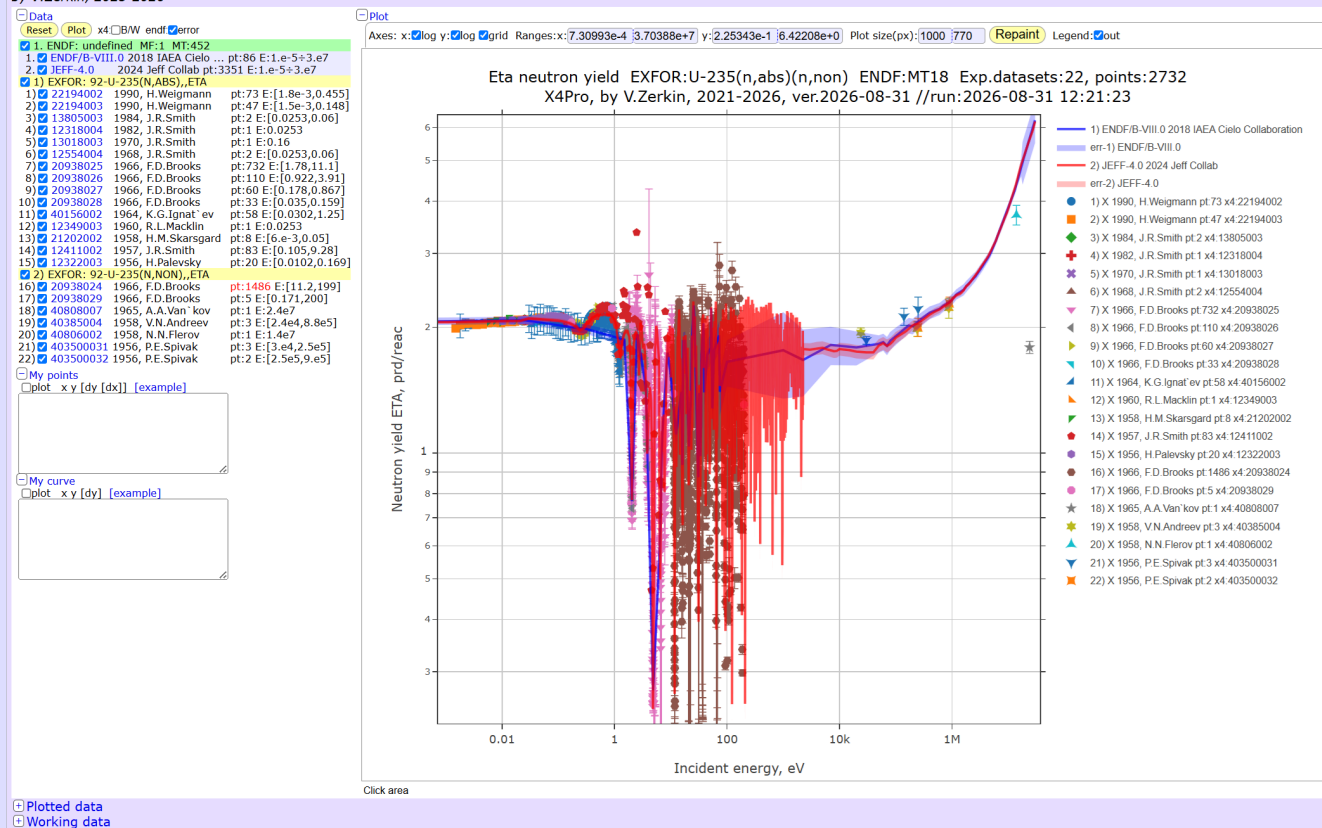
Reaction:92-U-235(N,ABS),,ETA Quantity:NU:y=Data(En) Datasets:22 Points:2732
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-10-02 13:52:27



part2-A-eta/ EXFOR/ENDF Eta: neutron yield per nonelastic even

18) part2-A-eta/out00/U235-eta.html.png ETA: EXFOR vs. ENDF $\eta \pm \Delta\eta$ 

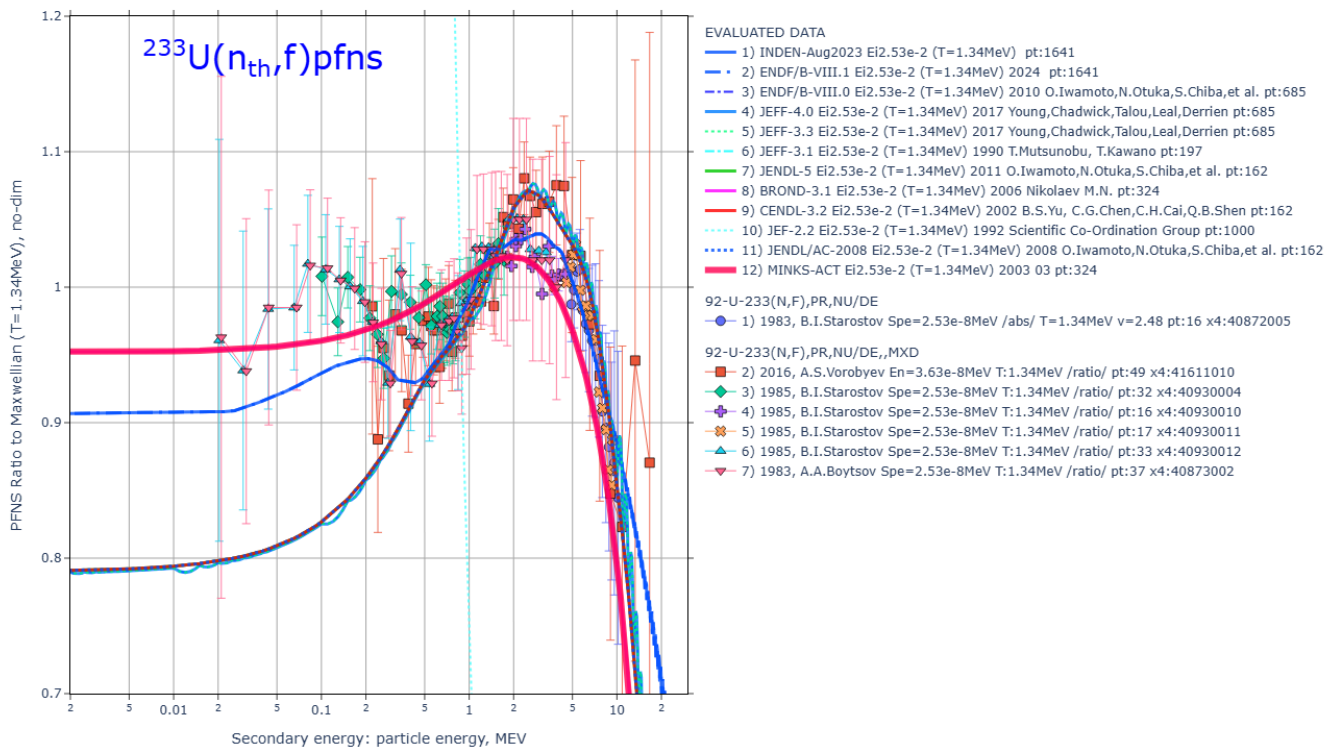
19) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT:452,18,102,MF:1,3,31,33

X4Pro - advanced plot
by V.Zerkin, 2025-2026

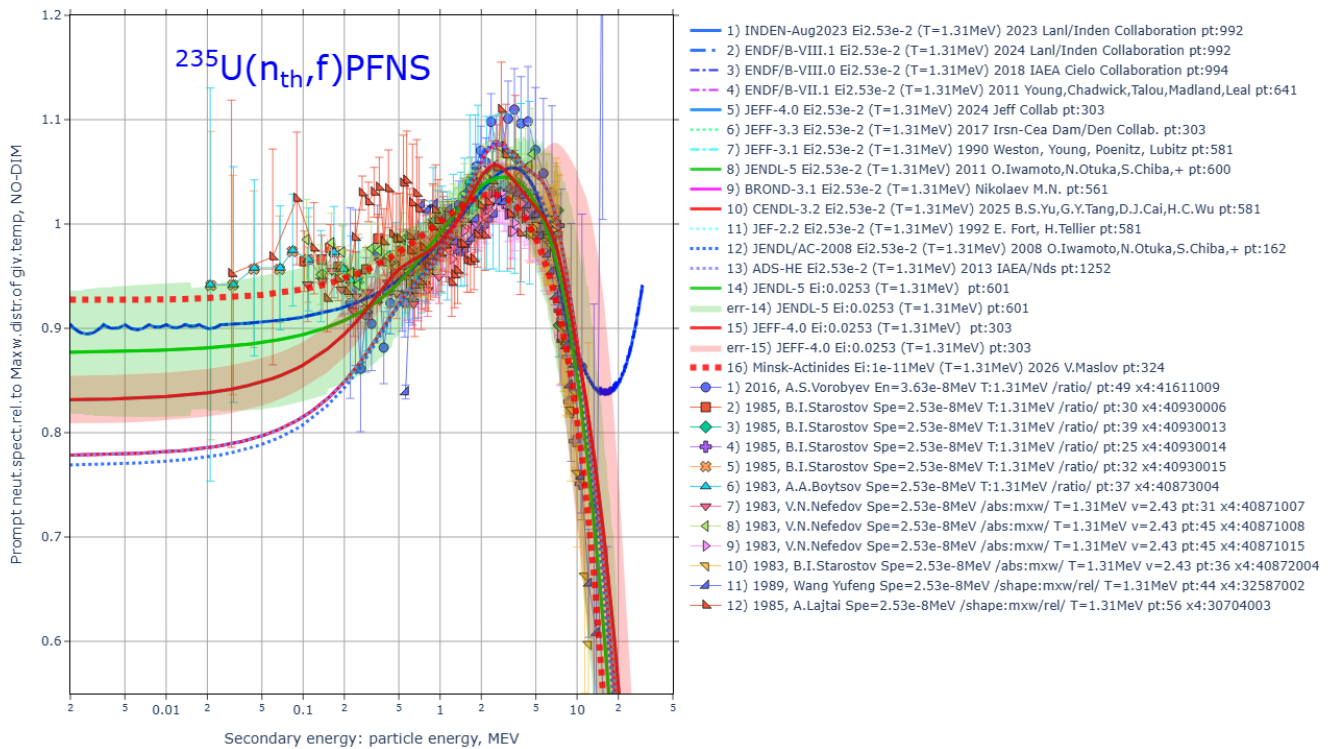
part2-B-pfns/ EXFOR/ENDF PFNS - energy spectrum of prompt fission neutrons

20) part2-B-pfns/out00/pfns233u00_0th.html.png $^{233}\text{U}(n_{\text{thermal}},f)$ PFNS-ratio to Mxw. (T=1.34MeV)

Reaction:92-U-233(N,F),PR,NU/DE Quantity:MFQ:y=Data(Spe,E2) Datasets:7 Points:200
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:54:58

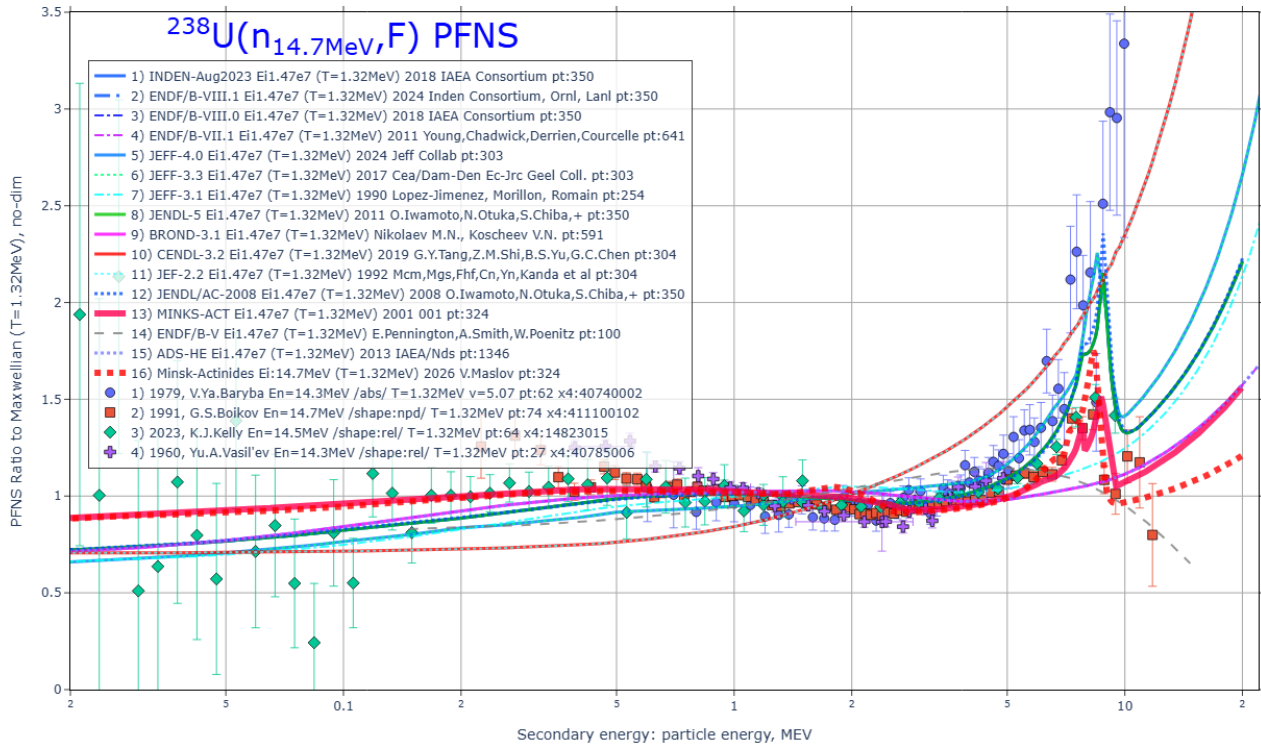
21) part2-B-pfns/out00/pfns235u00_0th.html.png $^{235}\text{U}(n_{\text{thermal}},f)$ PFNS-ratio to Mxw. (T=1.32MeV)

Reaction:92-U-235(N,F),PR,NU/DE,,MXD Quantity:MFQ:y=Data(En,E2) Datasets:12 Points:469
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:05



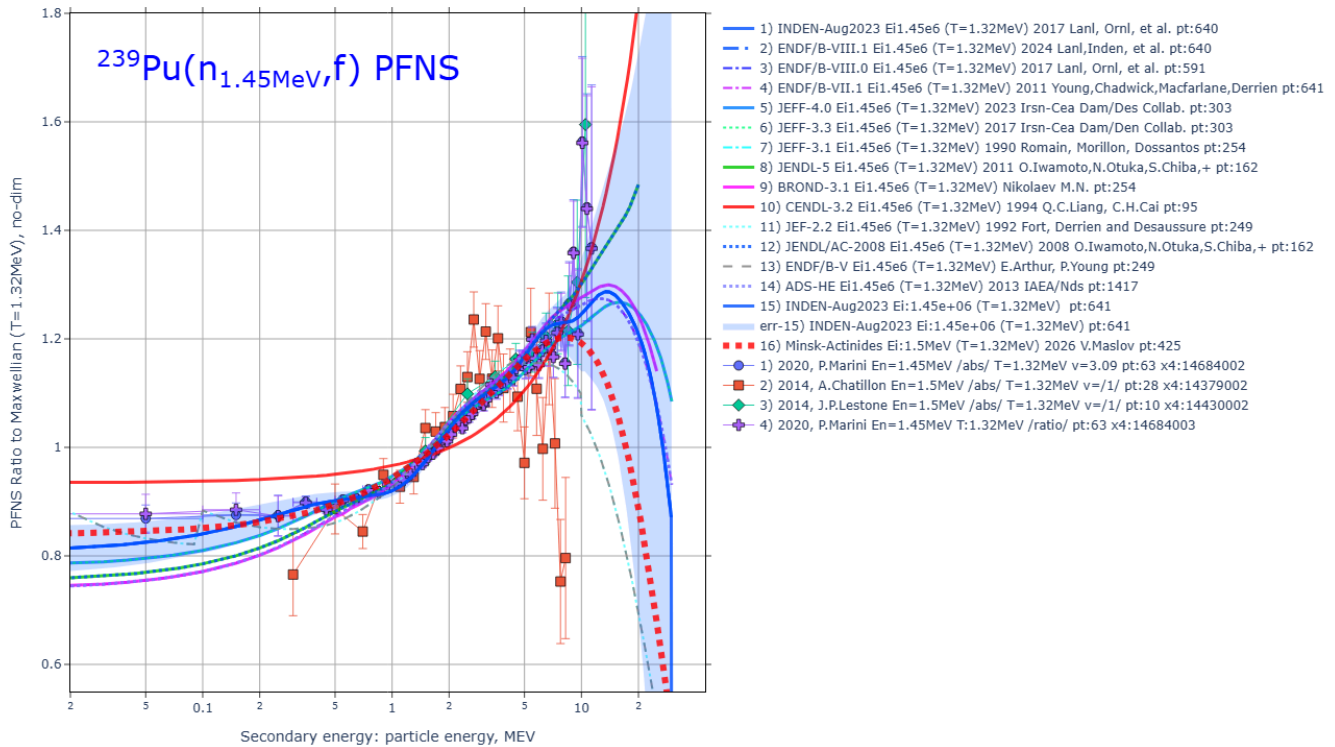
22) part2-B-pfns/out00/pfns238u14_7.html.png $^{238}\text{U}(n_{14.7\text{MeV}},f)$ PFNS-ratio to Mxw. (T=1.32MeV)

Reaction: 92-U-238(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 4 Points: 227
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:27



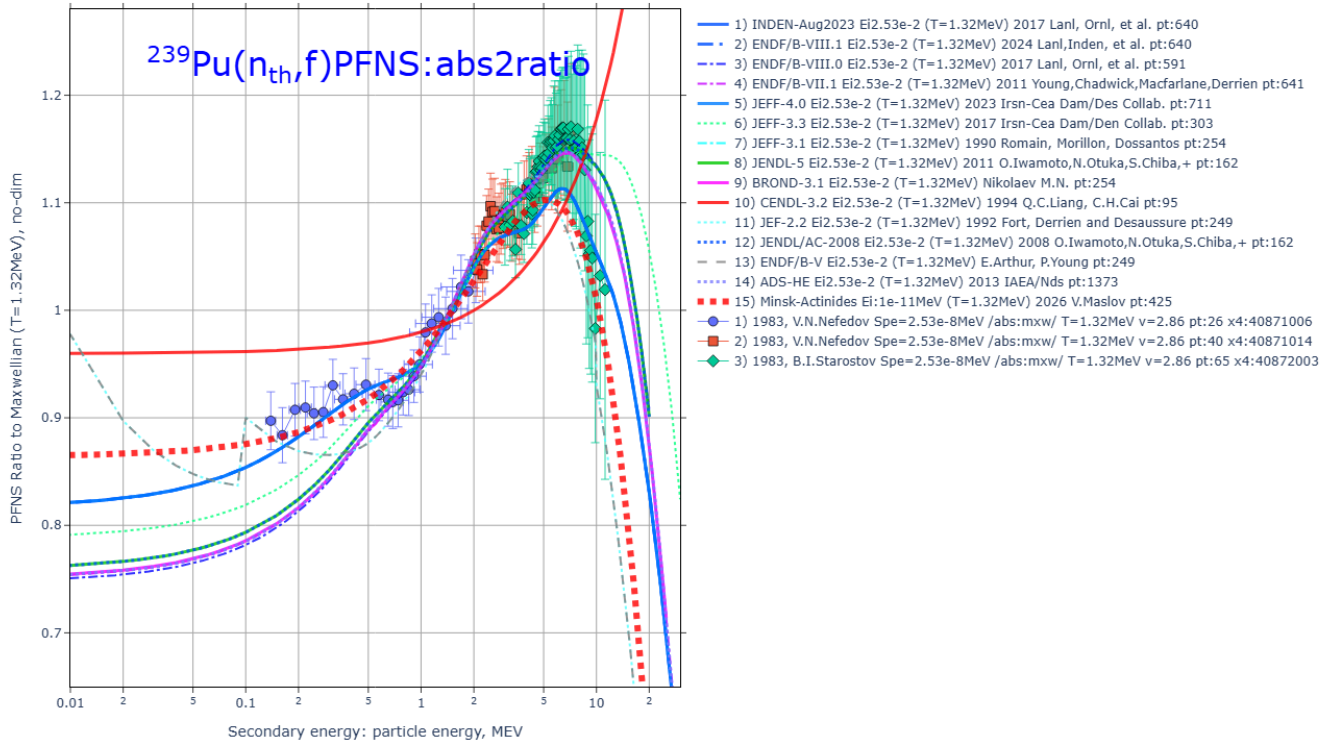
23) part2-B-pfns/out00/pfns239pu01_45.html.png $^{239}\text{Pu}(n_{1.45\text{MeV}},f)$ PFNS-ratio to Mxw. (T=1.32MeV)

Reaction: 94-PU-239(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 4 Points: 164/165
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:37



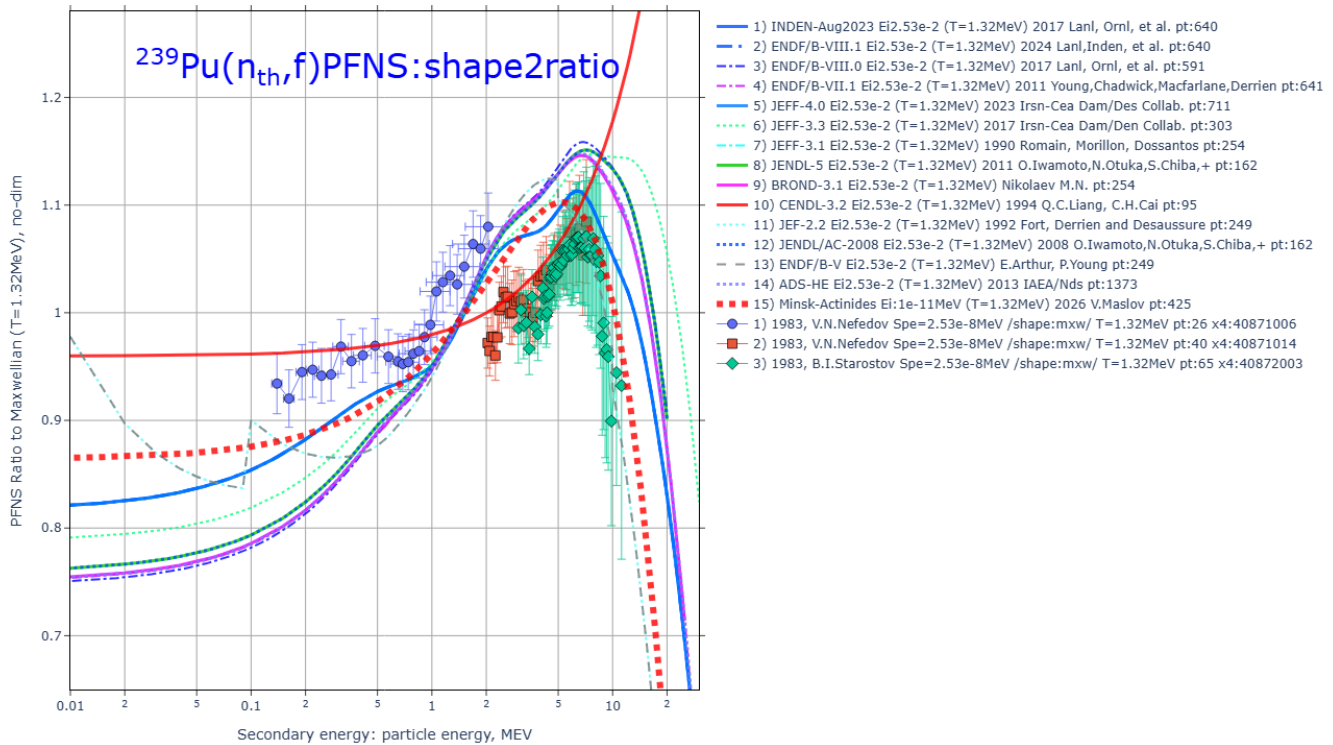
24) part2-B-pfns/out00/pfns239pu_t_abs.html.png $^{239}\text{Pu}(n_{\text{thermal}},f)\text{PFNS:abs2ratio}$

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:56:16



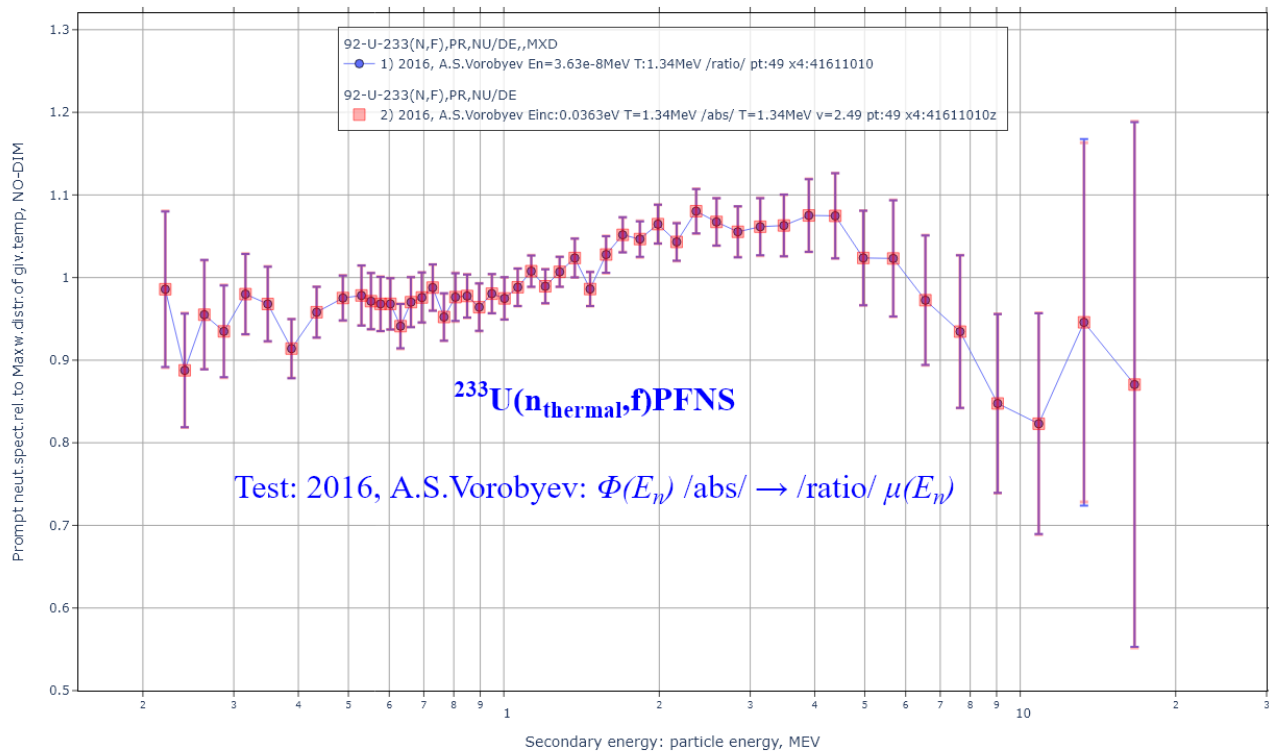
25) part2-B-pfns/out00/pfns239pu_t_shp.html.png $^{239}\text{Pu}(n_{\text{thermal}},f)\text{PFNS:shape2ratio}$

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:56:20



26) part2-B-pfns/out00/zabs2mxw233u_t.html.png [Check converting of abs.values to Mxw.ratio](#)

Reaction: 92-U-233(N,F),PR,NU/DE,,MXD Quantity:MFQ:y=Data(En,E2) Datasets:2/1 Points:98/49
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-20 //running:2026-09-22 17:58:10



27) fig12-vs-x4pro.png ²³⁵U(n_{15MeV},f) PFNS-ratio to Mxw. (T=1.32MeV)

PHYSICAL REVIEW C **111**, 044611 (2025)

Prompt-fission neutron spectra in the ²³⁵U(n, f) reaction

B. Mauss^{1,2,*} J. Täheb^{1,2} B. Laurent^{1,2} G. Bélier^{1,2} A. Chatillon^{1,2} D. Étasse³ P. Morfouace^{1,2} O. Roig^{1,2}

M. Devlin⁴ R. C. Haight⁴ and K. J. Kelly⁴

¹CEA, DAM, DIF, F-91297 Arpajon, France

²Université Paris-Saclay, CEA, LMCE, F-91680 Bruyères-le-Châtel, France

³LPC Caen, ENSICAEN, Université de Caen, CNRS/IN2P3, F-Caen, France

⁴P-3, Los Alamos National Laboratory, Los Alamos, New Mexico 87545, USA

X4Pro - PFNS

Reaction: 92-U-235(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:2 Points:121
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-24 //running:2026-09-25 00:45:39

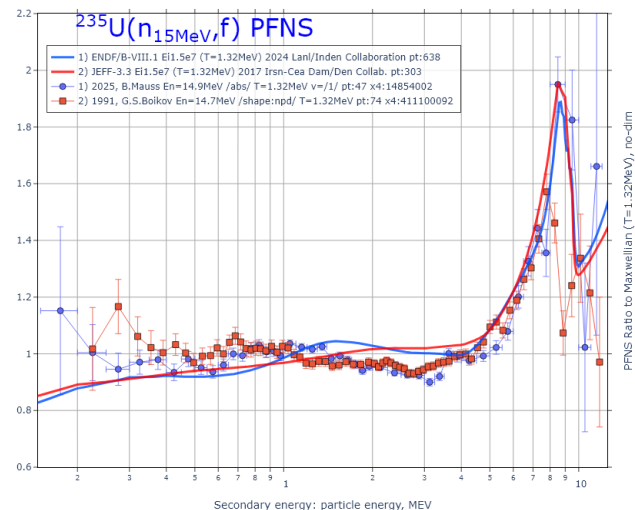
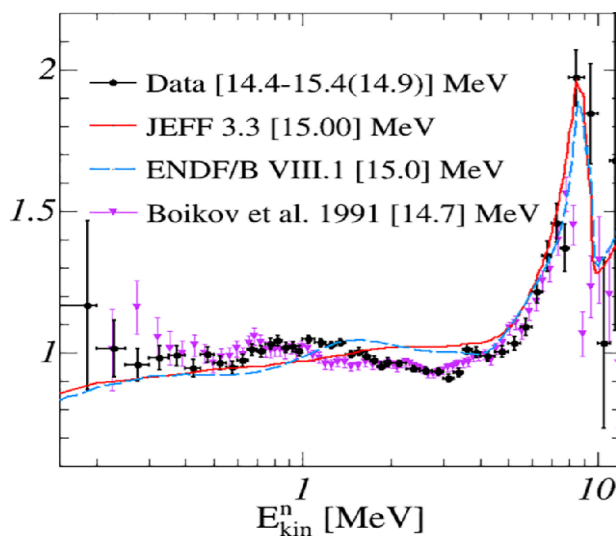


FIG. 12. Ratio of measured PFNS by a 1.32 MeV Maxwellian distribution for four incoming neutron energies. The horizontal bars give the bin size while the points are placed at the mean energy position of the energy distribution within each bin. The JEFF3.3 and ENDF/B VIII.1 evaluations are plotted for comparison. Data from Refs. [4,29,30] were also added for comparison. The corresponding energies are given in the legend.